Should You Pay to Remember? One Number, a Pre-Registered Protocol, and a Field Record for Pricing Regime Retention in Trading Systems

Anonymous Author(s)

Abstract

Regime-aware trading systems carry an evaluation problem that is worse than overfitting: their evaluations cannot fail. A regime-conditioned learner happily trains, "specializes," and beats a weak baseline on structureless labels, so a backtest win licenses nothing. We propose and field-test a pre-deployment protocol built around one number: the *retention deficit* $\hat{\Gamma}$, the post-reactivation decision-regret difference between the deployed always-retrain learner and an identical twin whose per-regime parameters are protected during dormancy—a two-arm counterfactual any desk can run before buying retention machinery. A structure screen the protocol once gated on is demoted by its own record—three dissociations from realized value—and survives only as a negative control. The field record spans daily crypto and commodities (screen fails, evaluations go flat, as required), 4-hour crypto (screen passes, $\hat{\Gamma} = 0$, flat again), and daily U.S. industry portfolios 1926–2025, where $\hat{\Gamma}$'s sign tracked the outcome direction in both directions: a positive deficit ($+8.9 \pm 1.1$ basis points of daily regret per probe day) coincides with the full pre-registered arm ordering, a negative one with an inversion, and a frozen-specification replication on a withheld 64-year era scores 5 of 6 cells. Crisis-window transaction costs amplify rather than kill the deficit ($+12.6 \rightarrow +23.0$ bps as cost tiers sweep 25–100 bps). An externally lodged, announcement-lagged NBER-recession test—registered with the OSF before recession dates ever touched the panel—resolved as a hit, with its two-recession concentration disclosed. A break-even rule converts $\hat{\Gamma}$ into a dollar-denominated carrying-cost threshold. All claims are regret and retention claims, not alpha claims, and the record's dependence structure is stated at the same prominence as its hits: its evidential force is closer to two-to-three effectively independent observations than to its twelve scored cells. A live temporal test (diagnostics frozen 2026-06-30, scored 2027-04-07) and a constituent-level replication are lodged in advance.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Machine learning*.

Keywords

regime switching, continual learning, decision-focused learning, evaluation protocols, pre-registration, transaction costs

1 Introduction: the evaluation that cannot fail

Every multi-strategy desk faces the same quiet question: the crisis playbook has been idle for six years—who is paying to keep it calibrated, and is that money buying anything? The machine-learning literature now offers an inventory of retention machinery to answer "yes": per-regime expert libraries, replay buffers, parameter isolation, continual-learning regularizers ported to finance [16, 28, 29, 31, 33]. What it does not offer is a way to find out that the answer is "no."

The reason is structural. A regime-aware evaluation has more researcher degrees of freedom than an ordinary backtest—the labeler, its thresholds, the dormancy definition, the evaluation windows—and its machinery produces organized-looking behavior on any labels at all: a regime-conditioned learner trains, converges, and "specializes" whether or not the labels carry information. Beating a generic baseline is therefore not evidence. *A method whose evaluation cannot go flat on structureless data should not be trusted when it goes sharp.* This paper is organized around engineering evaluations that can go flat, running them on real substrates, and reporting the flats, the inversions, and the hits in one ledger.

We work in the register of decision regret, not P&L: the per-day gap between the portfolio chosen from a model's predictions and the portfolio an oracle would have chosen, the predict-then-optimize lesson that prediction error and decision quality dissociate [8, 22]. The question the protocol prices is narrow and universal: when a dormant market regime returns, does a capacity-bounded learner that spent the dormancy retraining on whatever was active pay a measurable penalty in the first days of the reactivation—the days when decisions matter most—relative to a twin that kept the dormant regime's parameters untouched? If yes, retention is worth money and the break-even arithmetic says how much; if no, every dollar of retention machinery is premium paid on a claim that will never be filed. Recent theory makes the same point from first principles: a bounded agent should not retain what it can cheaply relearn [17]; our contribution is the measurement that decides the question per substrate, before deployment.

Contributions. (1) *A pre-deployment protocol* for regime-aware trading systems: a cheap falsification screen run first, a binding deficit diagnostic run second, and a break-even carrying-cost rule that converts the diagnostic into dollars (Section 2). (2) *The diagnostic itself*: $\hat{\Gamma}$, a two-arm counterfactual (deployed learner versus its dormancy-protected twin) measured in post-reactivation decision regret—one number, causal by construction, cheap enough to pre-register and run on any book. (3) *A resolved field record* (Section 3): seven substrates spanning crypto, commodities, and a century of daily U.S. equity industry data, every null and inversion included, with the screen's honest demotion, a costs battery in which the deficit strengthens net of crisis-window costs (Section 4), an externally custodied forward test that has since resolved, and a live temporal test whose diagnostics are already frozen (Section 5). The retention mechanism, its theory, and the controlled dissociation studies are developed in a companion manuscript [2]; this paper

ICAIF '26, Milan, Italy
2026. ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

is the protocol, the number, and the record a practitioner needs to decide whether to pay to remember.

1.1 Position among existing practice

Three literatures meet at this question and each leaves it unpriced. *Regime-switching finance*—Markov-switching models [12], regime effects in returns [1], regime-conditioned allocation [11, 26, 27], regime-weighted forecast combination [7]—conditions a model or its combination weights on an inferred state, implicitly assuming every component remains fully estimated while its state is dormant. That assumption is exactly what $\hat{\Gamma}$ measures the failure of under bounded capacity. *Continual learning in finance* makes the opposite unconditional bet: retention machinery is deployed because forgetting is presumed harmful [16, 28, 29, 31, 33], with evaluation by averaged forgetting metrics [21] rather than by a pre-deployment test of whether the substrate offers anything to retain. Our record contains real substrates where that bet loses ($\hat{\Gamma} < 0$ with a realized inversion), which is why the protocol runs *before* the purchase. *The backtest-overfitting protocol literature*—deflated Sharpe ratios [3], SPA and Reality-Check family tests [13, 34], the broader discipline of [14, 20]—is our closest methodological ancestor: it converts an unfalsifiable claim ("my strategy works") into a testable one by charging for selection. Our protocol does the same for a different claim ("my system needs to remember"), and inherits that literature's stance: the deliverable is the test, and the test must be able to say no.

Desks already answer the retention question, informally: shadow books and paper-traded dormant strategies, champion–challenger frameworks, and the scheduled revalidation cadence of model-risk management (in the U.S., supervisory guidance SR 11-7). What none of these supplies is a pre-committed falsification standard or a price: a shadow book with no lodged prediction can always be renarrated after the fact, and a revalidation calendar says *when* to look, not *what number decides.* The protocol below is the pre-registered, dollar-denominated version of the shadow book a desk already runs.

2 The protocol

2.1 Decision regret and probe windows

A stream of trading periods presents per-asset features X_t and next-period returns y_t, with a causal regime label r_t computed from information through $t-1$ only. The decision layer is a cardinality-constrained long-only portfolio (top-k by predicted return, position cap $w_{\max}$), and the per-period *decision regret* of a prediction $\hat{y}$ is $\rho(\hat{y}; y) = y^\top z^*(y) - y^\top z(\hat{y}) \geq 0$: the return given up relative to the oracle portfolio. Regret, not prediction error, is the protocol's metric throughout—errors that do not change the selected book are free, and errors at the selection boundary are not [8]. We quote $\hat{\Gamma}$ and all regrets in *basis points of daily return per probe day* (1 bp = 10^{-4}); transaction-cost *tiers* (25–100 bps) are bps of notional traded, a different object.

The model card. Every real-data number in this paper comes from one fixed, deliberately modest learner so that $\hat{\Gamma}$ is a property of the substrate, not of architecture search: linear per-regime heads on five causal cross-sectional features per asset (5- and 20-day momentum, 20-day volatility, 50-day moving-average gap, constant; cross-sectionally standardized), trained online by SGD on the SPO+ decision surrogate [8], under a hard capacity bound of K=2 heads for a four-regime taxonomy—capacity scarcity is the phenomenon under test, not an implementation accident. The portfolio layer is top-k=5 of ∼49 assets at position cap $w_{\max}$=0.2. Code, result files, and the manifest-driven audit that checks every quoted number against them are released.

Regimes recur with dormancy: the crisis label may sleep for years. The protocol's unit of account is the *probe window*: the first $N_p = 15$ trading days of every regime block whose regime was dormant at least 90 days. Post-reactivation regret—mean daily regret inside probe windows—is the primary metric because the reactivation transient is where retention could pay and where a fresh learner is most exposed. Neither constant is load-bearing: a lodged sensitivity battery (a 10-seed re-read of shared trajectories) varying $N_p \in \{5, 10, 15, 30\}$ and the dormancy cutoff over $\{60, 90, 120\}$ days leaves the deficit's sign positive and significant in every cell; the per-day deficit concentrates early (+13.4 bps at N_p=5 declining to +5.7 at N_p=30) while the cumulative harvest per reactivation saturates by $N_p \approx 15$–30.

Conventions, fixed once. Every comparison in this paper is walk-forward with causal features; every claim is a pre-registered ordering prediction tested across seeds (100 where stated, else 20) with Welch tests and Holm–Bonferroni correction inside each registered pair family; paired designs expose every arm to identical data streams per seed. Where a design's seeds vary only initialization—the walk-forward batteries—the resulting intervals are implementation noise conditional on one market history, and every table says so. Refuted predictions are reported at the same prominence as confirmed ones; this paper contains both.

2.2 The structure screen, honestly demoted

Screen (structure). Fit one model per regime label, walk-forward; compare its decision regret against the identical machinery run on block-shuffled labels (≥ 50 draws), summarized as a z-score against the shuffle distribution (the empirical permutation p cannot fall below 1/51; a normal fit to the control distribution gives the indicative p), with a split-half stability check and a comparison against a pooled unconditioned model. Block-shuffling preserves label persistence while destroying the label-return association, so the control inherits the machinery's ability to look organized on nothing. The screen asks only: *do these labels carry decision-relevant information at all?* It costs minutes and requires no retention machinery.

We designed the screen as a gate—no pass, no claims—and its own field record demoted it. Across the substrates of Section 3 the screen's verdict dissociated from realized retention value three times: a passing screen preceded an inverted outcome (daily equities under a volatility-band labeler), the strongest pass we ever measured (z = 23.5, withheld 1926–1989 era) coincided with an inversion, and a failing screen preceded a positive, cost-robust deficit (the NBER labeler). The screen therefore *licenses nothing.* It survives in the protocol for its original falsification role—where it fails, a null evaluation is the predicted outcome and a sharp one is suspect—and as a cheap negative control: machinery that "wins" where the screen fails is fitting noise.

2.3 The deficit $\hat{\Gamma}$: one number from two arms

Diagnostic (binding). Run two arms of identical capacity, features, and learning rates over the same walk-forward stream:

- **A1 (deployed default)**: a capacity-bounded learner that always trains on the currently active regime—the honest model of any allocation policy that follows current reward, whether a retraining schedule, a learned router, or recent-performance capital weighting;
- **A9 (protected twin)**: the same learner with per-regime parameters pinned—each regime's parameters are trained only on that regime's data and are untouched during its dormancy.

The *retention deficit* is $\hat{\Gamma} = \overline{\rho}_{\mathrm{A1}} - \overline{\rho}_{\mathrm{A9}}$ over probe windows: what dormancy-period retraining actually cost, in regret per probe day. It descends from the econometric post-break estimation-window question—how much pre-break knowledge should a forecaster retain [30]. It is causal by construction (the arms differ in exactly one design bit), cheap (two arms, no mechanism), and signed: $\hat{\Gamma} \leq 0$ is a finding, not a failure. Where $\hat{\Gamma}$ is positive, any isolation-preserving implementation collects it; where it is not, no retention mechanism can be bought, because there is nothing to collect.

Both arms are stand-ins for classes, which is what makes the two-arm run sufficient—and the boundary between the classes was itself pre-registered and tested. A1 represents *recency-driven* allocation: every policy whose training data or capital tracks current conditions—a rolling-window retraining schedule, a learned router over experts, recent-performance capital weighting across desks—all of which starve the dormant regime, because a dormant regime emits no reward to follow. The most common desk default, *expanding-window* retraining on all history, is **not** in A1's class, and we did not assume the answer: an expanding-window arm was lodged with branch predictions and run; it collected 73% of the deficit (the lodged modal branch, ≥80%, missed low; residual versus the protected twin $+2.5 \pm 2.5$ bps, not significant at 20 seeds), behaving as what it is—data retention without parameter isolation. The narrowed class is still the modal one in systematic allocation: rolling-window retrains, learned routers, and performance-weighted capital are the default shapes of multi-strategy platforms. A9 represents every *isolation-preserving* implementation: a siloed desk, a frozen per-regime adapter store, an expert library with pinned members—and, at the cheap end, data retention alone: replaying stored crisis episodes collects roughly 99% of the deficit, expanding-window training most but not provably all of it. If your deployed policy already retains all history, $\hat{\Gamma}$ prices your residual exposure as small on this substrate; if it is recency-driven in data or in capital, $\hat{\Gamma}$ prices the full deficit. The protocol prices the class boundary, not any implementation; choosing an implementation is a separate decision the companion studies [2]. Two practical notes. First, $\hat{\Gamma}$ is measured net of *rehearsals*: if near-miss episodes keep the dormant specialist warm between major events, that recalibration is in the measurement, which is the economically relevant accounting. Second, $\hat{\Gamma}$ should be re-measured net of transaction costs before pricing (Section 4); in our record costs move the number, in one case across zero.

2.4 Pricing the deficit: the break-even rule

$\hat{\Gamma}$ is in return units per probe day, so it converts directly to dollars. Charging a carrying cost c (dollars per idle trading day: infrastructure, data, attention, the calibration operation) against a sleeve of notional value V, retention of a regime whose dormancy is D trading days pays over one dormancy–reactivation cycle iff

$$\boxed{c \cdot D \;\leq\; \hat{\Gamma}_{\mathrm{net}} \cdot N_p \cdot V}$$

where $\hat{\Gamma}_{\mathrm{net}}$ is the deficit measured net of transaction costs (Section 4) and N_p is the probe length. The rule is a special case of the break-even analysis developed in the companion [2]; here it is used only as arithmetic.

Worked example (a \$100M crisis sleeve), priced as a decision under uncertainty. The measured net-of-cost deficit at the 25 bps crisis-cost tier is $\hat{\Gamma}_{\mathrm{net}} = 12.6 \pm 2.0$ bps per probe day (Section 4); on \$100M the point estimate is \$126,000 per probe day, ≈\$1.89M of avoidable regret per reactivation ($N_p = 15$). A committee should not price off the point: the interval is implementation noise conditional on one market history, so the defensible input is the CI lower bound, further discounted by how little the committee's sleeve resembles the book the number was measured on (a top-5-of-49 long-only industry portfolio). Table 1 prices the rule across the interval and across the record's own deep-dormancy draws rather than the single June-2020 draw:

Table 1: Break-even carrying cost (\$K/year on a \$100M sleeve, 252 trading days) across the $\hat{\Gamma}_{\mathrm{net}}$ 95% CI and the record's deep-dormancy range pooled across labelers (D=1,378 and 2,445 are NBER-cell entries; 2,953 is the drawdown cell's maximum). Retention pays below the number; a \$1M/year staffed idle desk pays nowhere in this table, a \$50K/year frozen-parameter calibration job pays everywhere.

	D=1,378	D=2,445	D=2,953
$\hat{\Gamma}_{\mathrm{net}}$ CI-low (10.6 bps)	291	164	136
point (12.6 bps)	346	195	161
CI-high (14.6 bps)	400	226	187

The decision is interval-stable here—the *order of magnitude* separates a staffed desk from a calibration job at every cell—and that stability, not the point estimate, is what the rule should be trusted for. Nor is the harvest an artifact of the probe constant: in the lodged sensitivity battery the cumulative (gross) avoidable regret per reactivation is ∼165–172 bps-days whether measured at N_p=15 or N_p=30. And a desk should price its own class: if the deployed default is expanding-window retraining, the relevant input is not the full deficit but the residual of Section 2.3 ($+2.5 \pm 2.5$ bps)—point estimate ≈\$39K/year break-even at D=2,445, interval [\$0, ∼\$77K]—below the staffed-desk line and possibly below the calibration job. The full deficit is the price of recency-driven allocation, not of every desk. The rule's two-sided honesty is the point: it prices some retention out, and that is the protocol working.

3 The field record

Table 2 is the paper's spine: every substrate the protocol has touched, with the screen verdict, the measured deficit, and what the ten-arm evaluation machine (the companion's arm family, spanning reward-driven and retaining allocations [2]) actually did. Conventions: all labels causal; all comparisons walk-forward; Welch tests with Holm correction inside each pre-registered family; 100 seeds where stated, else 20. One caveat governs every real-data row and is repeated in Section 6.2: each walk-forward window is a single market history, so intervals are seed noise conditional on that history, not sampling uncertainty across markets.

Daily crypto and commodities: the protocol's floor. Neither small daily panel shows decision-metric structure under either causal labeler (all $|z| < 0.7$): regime-conditioned regret is statistically identical to block-shuffled-label regret. The ten-arm evaluation run on regime-stitched crypto schedules goes flat exactly as the failed screen predicts (no pair survives Holm correction; min Holm $p = 0.17$), and the numerically best arm is the one that *learns nothing*—a fixed-strategy combination—which sharpens the null: where labels carry no structure, machinery that adapts to them buys less than machinery that ignores them. This is the "can go flat" property demonstrated, not asserted.

4-hour crypto: structure without a deficit. A richer intraday feature inventory locates the program's first screen pass (951 episodes; conditioning beats shuffled labels by 3.15%, $z = 2.32$, both halves positive)—with its selection caveats stated at the same prominence: it is one confirmed cell from a ten-cell screen, and the confirmation reused the screen's data span, so the gap estimate is biased upward by selection on the maximum. The evaluation is flat anyway—and the two-arm counterfactual explains why retrospectively: $\hat{\Gamma} = +0.02$ bps, zero at the record's precision. Reactivations are elevated for every arm including the protected twin; regime onsets are intrinsically hard here, but nothing is forgotten—at this bar density a fresh learner refits within the probe window. The epistemic order, plainly: the lodged prediction for this cell was separation; it was refuted; the zero deficit is a post-hoc diagnosis whose honest test is every substrate scored since. This cell is why the deficit, not the screen, is the binding diagnostic: structure is necessary for anything to be learnable, and irrelevant to whether retention pays.

Daily equities, 1990–2025: the deficit exists, and its sign prices the pool. On the Ken French 49 industry portfolios (daily, value-weighted, 1990–2025; 9,067 days), with designs and predictions pre-registered before each run, two labelers deliver the two halves of the story. Under the volatility-band labeler (L1) the screen passes and the deficit is *negative* (-1.3 ± 0.7 bps): the machine run anyway produces a significant inversion—the always-retrain baseline beats the retaining arms in the probe window (Holm $p < 10^{-6}$), retention premium paid with no claim to collect. Under an index-drawdown labeler (L3, 15% threshold fixed in the registration before any run) the screen *fails* by the letter of its pooled sub-criterion while the deficit is positive—the first positive $\hat{\Gamma}$ on any real substrate in this program ($+8.9 \pm 1.1$ bps, 100 seeds)—and the full pre-registered arm ordering emerges, every pair of the chain below the displayed cap (Holm $p < 10^{-6}$; exact values, spanning 10^{-15} to 10^{-65}, are in the released files), with the retaining arms also ahead on overall regret. The ordering survives the strongest desk default as well: against the pre-registered expanding-window arm—all history retained, no isolation—the retaining pool with the invariance objective stays ahead (Holm 1.8×10^{-3}). One labeler-anatomy fact matters for reading the number: the *union* of the drawdown labeler's crisis states recurs every one to three years (the 2010, 2011, 2015–16, and 2018 near-misses are all in the inventory), while the specific crisis-with-uptrend cell sleeps up to 2,953 trading days—so retained parameters do receive live-fire recalibration between major crises, and $\hat{\Gamma}$ is measured net of those rehearsals, as a desk's economics would be. The deficit's evaluated inventory is 27 reactivations; under leave-one-reactivation-out it survives every single-event exclusion (range [7.1, 10.1] bps, all significant), survives excluding all of calendar-2020 ($+8.0 \pm 1.9$ bps), and is independently positive within the 2010s ($+7.7 \pm 2.2$ bps, n=14) and 2020s ($+9.4 \pm 4.0$ bps, n=12).

Three disclosures belong beside that result. First, *threshold fragility*: the deficit lives at essentially one drawdown threshold per era—at 10/12/20% on 1990–2025 it is null or negative ($+1.3 \pm 2.7$, -1.0 ± 1.7, -4.5 ± 2.6 bps)—and the pre-registered robustness prediction was refuted. Second, a *favorable wrinkle we flag because it was not predicted*: at 100 seeds the schedule-resampled (stitched) counterparts, previously flat, turn marginally positive in $\hat{\Gamma}$ under both labelers ($+2.3 \pm 1.6$ and $+2.3 \pm 1.5$ bps) while their arm families stay flat—consistent either with single-history optimism in the walk-forward precision or with stitching destroying exactly the multi-year dormancy structure that makes retention pay; the lodged constituent-level test discriminates. Third, *mechanism honesty*: registered probes found the deficit does not decay across the probe window and follows no dormancy or rehearsal covariate; it behaves as a persistent regime-conditional allocation deficit—the reward-following learner serves crisis windows persistently worse, the learner-side analogue of slow-moving capital [6, 24]—not demonstrated eviction-forgetting, which remains a controlled-setting result in the companion [2]. The deployment consequence (the counterfactual prices retention either way) is unchanged; the mechanism attribution is open, and we say so.

The withheld era, 1926–1989. The byte-identical frozen specification—same features, arms, tests, and the lodged sign hypothesis; zero new researcher degrees of freedom—was replicated on 36 industries over 16,983 days the design had never seen, in the pre-discovery-sample tradition [9, 18]. The screen delivers its strongest pass ever ($z = 23.5$) while the frozen 15% cell *inverts* ($\hat{\Gamma} = -2.4 \pm 0.8$ bps; the always-retrain baseline wins, Holm 2.1×10^{-3})—the dissociation that finally demoted the screen. The deficit's sign, by contrast, scored: of six withheld cells (thresholds 10/12/15/20%; two sub-eras at 15%), five are consistent with the lodged sign rule, in both directions, including two positive-deficit cells reproducing the full ordering out-of-sample (10%: $\hat{\Gamma} = +2.0 \pm 0.9$ bps, Holm 1.9×10^{-6}; 1958–89: $\hat{\Gamma} = +2.1 \pm 1.5$ bps, Holm 2.1×10^{-4}) and two negative-deficit cells with the predicted inversion. The single miss (12%: $\hat{\Gamma}$ negative with a CI excluding zero, arm table flat) is scored against the rule under its strong form and for it under its weak form; Section 5 carries the full accounting. The paying threshold *relocated* across eras (15% → 10%) as crisis density shifted—the fragility replicates in pattern, not in location,

Table 2: The field record: substrate × (structure screen, deficit $\hat{\Gamma}$, evaluation outcome). $\hat{\Gamma}$ is in bps of daily regret per probe day (A1–A9, probe 15 days, dormancy ≥ 90 days, bar-native cutoff on the 4-hour cell; ± is a 95% CI over seeds, conditional on each window's single history). A screen can fail its registered pooled sub-criterion while its shuffle-z is positive (Section 3). The evaluation family behind "what the machine did": the deployed always-retrain baseline (A1), its dormancy-protected twin (A9), specialist pools with and without an invariance penalty, a learned router, recent-performance and hedge-style capital weighting, replay and banded variants. "Ordering" = the pre-registered arm ordering (retaining arms ahead of reward-driven arms); "inversion" = the deployed always-retrain baseline wins; "predates $\hat{\Gamma}$" = the battery ran before the deficit diagnostic entered the protocol. Displayed p-values are capped at 10^{-6}; exact values are in the released result files, and every row's result files are released.

Substrate (span; labeler)	Structure screen	Deficit $\hat{\Gamma}$ (A1–A9)	What the machine did
Crypto, daily (5 pairs, 2021–25; vol/trend + filtered-vol)	fail ($z = -0.66$ / -0.17)	— (predates $\hat{\Gamma}$)	flat, as predicted (min Holm $p = 0.17$; 100 seeds)
Commodities, daily (3 series, 2015–25; both labelers)	fail ($z = -0.14$ / $+0.16$)	— (predates $\hat{\Gamma}$)	no dissociation run; claims forbidden by screen
Crypto, 4-hour (bar-native filtered-vol; confirmed cell)	pass (+3.15%, $z = 2.32$)	+0.02 bps (≈0)	flat ($p > 0.57$ all pairs): structure, nothing to retain
French 49 industries, daily (1990–2025; L1 vol-band)	pass (+1.14%, $z = 3.18$)	−1.3 ± 0.7 bps (100 s)	*inversion*: always-retrain wins (Holm $p < 10^{-6}$); reverses net of costs
French 49 industries, daily (1990–2025; L3 drawdown@15%)	fail (registered pooled criterion; $z = 2.24$)	+8.9 ± 1.1 bps (100 s)	full ordering (every chain pair Holm $p < 10^{-6}$); cost-amplified
Withheld era: 36 industries (1926–89; frozen L3 spec, 6 cells)	strongest pass ($z = 23.5$)	sign varies by cell	sign rule 5/6, both directions; screen's best pass → inversion
NBER recession × trend, announce-lagged (1990–2025)	fail ($z = 1.74$)	+8.1 ± 3.3 bps	ordering (Holm $1.6{\times}10^{-4}$); two-recession concentration disclosed

so any deployment must estimate the window's location for its own era.

The NBER cell: temporal custody. The one cell whose ex-ante status a third party can certify used a labeler with no free threshold anywhere: regime = NBER recession state × 50-day trend, the causal primary variant admitting a recession only from its *announcement* date. The registration was deposited with the Open Science Framework at 22:47 ET on 2026-07-15; NBER dates were first joined to the panel at 22:55 ET. Recession-cell dormancies are crisis-scale by construction (up to 2,557 trading days). The screen *fails* both variants ($z = 1.74$ primary)—the third dissociation of screen from value—while the deficit prices positive: $\hat{\Gamma} = +8.1 \pm 3.3$ bps, strengthening to +10.5 ± 3.3 bps net of 25 bps crisis-window costs, with the ordering exactly as the rule demands (Holm $1.6{\times}10^{-4}$ gross; the full retaining chain at Holm ≤ $3.1{\times}10^{-5}$ net). Disclosures at equal prominence: the lodged leave-one-out and era-blocked slices show the deficit survives every one of 20 single-event exclusions and all calendar exclusions (drop-2020: +4.2 ± 2.6 bps; drop-2008/09: +3.6 ± 3.0 bps), but its mass concentrates in the two deep-dormancy recession entries (January 2009, dormancy 1,378 days; June 2020, dormancy 2,445 days; dropping the former alone cuts $\hat{\Gamma}$ to 41% of headline), and the 2010s era block—which contains no deep-dormancy recession entry, only one shallow 140-day tail reactivation—is negative-significant. The cell is event-robust in the strict sense, two-recession-concentrated, and not era-consistent; it is materially weaker than the drawdown cell's robustness profile, and we say so rather than average it away.

4 Costs and economics

A retention effect that dies under transaction costs is a curiosity; this one strengthens. The pre-registered cost battery charges crisis-window effective costs of 25/50/100 bps under a both-pay convention (every arm, including the protected twin, charged its own turnover; an arm-only variant was lodged as sensitivity and matches to the digit, because the twin's cost term cancels in the regret difference).

The deficit is cost-amplified. On the positive-deficit equity cell, the net-of-cost deficit rises monotonically with the cost tier: $\hat{\Gamma}_{\text{net}} = +12.6 \pm 2.0$ bps at the 25 bps tier, +16.1 ± 2.1 at 50, +23.0 ± 2.4 at 100, with the four-arm ordering intact and the headline contrast below Holm $p < 10^{-6}$ at every tier, strengthening monotonically with the tier. The mechanism is turnover, and it is the economically interpretable one: a reactivated always-retrain learner churns its top-k book hardest exactly when spreads are widest (measured turnover 1.03/day for the deployed baseline versus 0.62 for the calmest retaining arm). Paying to remember buys, among other things, *not trading like a fresh learner during a crisis onset.* The same signature reproduced ex ante on the NBER cell (+8.1 → +10.5 bps net of the 25 bps tier; turnover 1.04 vs. 0.63)—a labeler the cost battery never saw.

The inversion was a gross-only phenomenon. Under the volatility-band labeler, where $\hat{\Gamma} < 0$ and the always-retrain baseline won gross, the inversion *reverses* net of costs at every tier (paired difference −7.5 ± 1.0 bps at the 25 bps tier, $p < 10^{-6}$): net of realistic crisis-window costs the retaining arm wins even where the parameter-level deficit is negative, because the turnover discipline alone is worth more than the deficit costs. We report this as economics, not vindication: the sign rule prices the *parameter* claim, and turnover discipline is in principle separately purchasable. One sensitivity is disclosed and not acted on: charging costs inside the structure screen would flip the drawdown labeler's pooled sub-criterion from −0.10% to +3.0%; the screen as registered is a gross-metric criterion and remains a fail.

The desk countermeasure was pre-registered and run: a banded baseline does not buy the retaining arms' edge. A no-trade hysteresis band on the deployed baseline—update the served book only when more than b names would change, $b \in \{1, 2\}$—is the cheapest desk implementation of turnover discipline, and its control battery was lodged with an adverse branch stated before the run (had banding closed the net gap, the net-superiority claim would have been withdrawn in favor of "any turnover-disciplined implementation collects it," at full prominence). The adverse branch is refuted. Banding is free—gross regret unharmed, net improved—yet the retaining arm's net advantage stands on both equity cells: on the positive-deficit (drawdown) cell it beats both banded baselines at Holm $p < 10^{-6}$ with 97%/79% of the net gap intact at b=1/b=2, and on the volatility-band cell a significant residual survives (+2.8 ± 1.1 bps, Holm 2.1×10^{-5}), with the disclosure that the b=2 band closes 62% of that cell's reversal magnitude—the reversal keeps its label with the attenuation reported alongside. One lodged magnitude failed and is disclosed: the bands cut the baseline's turnover only 4.1%/15.7% against a registered ≥30%, because the crisis churn comes from daily re-rankings deeper than two names, which defeat narrow bands. That failed magnitude also scopes the conclusion: narrow hysteresis bands—the cheapest desk discipline—do not buy the retaining arms' net edge, and the bands' weakness as an instrument means stronger turnover controls (a turnover-penalized optimizer, a trade-rate cap) remain untested; within what was tested, the net edge behaves as *selection quality in crisis windows*, not just churn discipline.

What the cost model omits is market impact, borrow, and capacity; every substrate outside the two equity batteries remains gross; and none of these numbers is a P&L claim (Section 6.2). One directional note on the omission: the measured mechanism is excess turnover concentrated at crisis onset, exactly where impact is most convex and a linear-bps schedule understates true cost—so on the amplification result, the linear model errs against us, not for us; capacity of the underlying book is a separate question the schedule cannot answer. For any future alpha claim the lodged protocol of record is deflated Sharpe [3] plus an SPA/Reality-Check family test [13, 34], per the backtest-overfitting literature [14, 20].

5 The sign rule: a lodged, once-resolved hypothesis

The record above compresses to one deployment rule: *the sign of* $\hat{\Gamma}$ *prices retention.* Positive deficit ⇒ the retaining arms collect it; zero or negative ⇒ retention pays premium without claims. This section states exactly how much evidence that rule has.

The confession first. When first formulated, the rule was postdictive—written after seeing both 1990–2025 outcomes, and sharing its A1 arm with the contrasts it prices (its non-circular content is that the protected twin sides with the retaining arms exactly when the deficit is positive). The weak/strong scoring taxonomy below was formalized only after the withheld-era runs, and the rule as lodged was scoped to walk-forward designs, but the stitched designs are counted against ourselves anyway. One further scoring choice: no registration fixed whether the gross or the net-of-cost register adjudicates the NBER cell's strong form—gross, its retaining chain is partial; net (the economically primary register by the cost battery's own argument, an argument that predates the cell), it is a full hit—and the amendment lodged for the constituent test fixes the net register in advance so the ambiguity cannot recur.

The inclusive scorecard. Scoring every cell the rule could touch—six withheld-era cells, the two schedule-resampled (stitched) 1990–2025 designs, and the four NBER cells—the record is **8/12 under the strong form** (sign predicts the arm-table direction: ordering when significantly positive, inversion when negative) and **12/12 under the weak form**, which is one-directional: dissociation *only if* $\hat{\Gamma}$ is significantly positive—no cell dissociates without a positive deficit; the converse fails in the two stitched cells, and those failures are counted above as strong-form misses. Read those counts with their dependence structure first: **the scorecard's evidential force is closer to two-to-three effectively independent observations, in both directions, than to twelve trials**—the withheld 15% full-era cell is the union of its two sub-era cells, the four threshold cells share one history's crises, and the stitched designs resample the same panels. The four strong-form misses share one shape: a marginally significant $\hat{\Gamma}$ with a flat arm table—which is the rule stating its own detection floor: at the record's per-event dispersion (~25 bps), deficits below roughly 5 bps per probe day sit where the arm table cannot resolve them, and should be read as unpriceable at current inventory, not as retention verdicts. The two 1990–2025 walk-forward tests that motivated the rule are not counted at all.

Custody. Because a rule of this shape is exactly the kind of thing a file drawer manufactures, every registration in the program is graded by the strength of its timestamp custody, against ourselves: the first-day batteries are *self-attested* (specification and results entered the repository together); the later batteries—costs, replay, mechanism probes, the withheld era, the banded and expanding-window controls with their lodged adverse branches—are *git-separated* (specification pushed publicly before any run); the NBER cell *resolved under third-party custody* (OSF deposit 22:47 ET, recession dates first joined 22:55 ET); all future registrations are OSF-first. The full ledger, with commit hashes and verification instructions, is released; the OSF project is anonymized for review [TODO: replace OSF ID with anonymized view-only link]. As of this record every *run* registration in the program is closed; open are the two market-clock forward tests—the temporal-deployment scoring (2027-04-07) and the CRSP constituent battery—and two lodged-but-unrun registrations we disclose at the prominence a file drawer would deny them: a granularity-window map and a regional battery with region-specific sign predictions, scheduled before any further reanalysis of the 1990–2025 panel.

The live test. The rule's standing changes register only with temporally split measurement, so the program's next cells are already committed. (1) *The deployment-form temporal test*: all diagnostics frozen **2026-06-30**, on the vendored panel through 2026-05-29 ($\hat{\Gamma}$ = +8.1 ± 3.3 bps on the NBER cell; +9.1 ± 2.0 bps on the drawdown cell—both positive, so the lodged prediction is that the retaining arms outperform in the outcome window), scored against realized reactivations in 2026-07-01 through 2027-03-31 on a pre-named date, **2027-04-07**, with a void declared (not a hit) if no qualifying reactivation occurs. One clean refuting cell falsifies the deployment form of the rule. (2) *Constituent-level replication*: the CRSP S&P 500 panel, registration lodged before any data access, with a T-split amendment fixing the net-cost register and the window estimator in advance—the test that decides whether the deficit survives disaggregation, the canonical worry for industry-level effects [10, 25]. Both are reported at headline prominence regardless of outcome.

6 Field kit and limitations

6.1 Running the protocol on your own book

(1) **Freeze the design.** Fix the labeler family, thresholds, probe length, dormancy cutoff, and test family in writing; lodge the document externally (OSF or equivalent) before any run. The record above is the argument that this step is load-bearing.
(2) **Label causally.** Regime labels must be computable from information through $t-1$—announcement dates, not revision dates, for any official indicator.
(3) **Run the screen.** One model per regime versus the same machinery on ≥ 50 block-shuffled label draws, on *your decision metric*, walk-forward, with a split-half check. A fail predicts your regime-aware evaluation should go flat; run it anyway and distrust everything if it does not.
(4) **Count your inventory, then measure $\hat{\Gamma}$.** Two arms: your deployed allocation policy, and its twin with per-regime parameters pinned through dormancy. $\hat{\Gamma}$ = probe-window regret difference. Charge your own crisis-window cost schedule (both-pay); keep $\hat{\Gamma}_{\text{net}}$. Power first: at our record's per-event dispersion (~25 bps across reactivations), an event-level test cannot resolve a ~9 bps deficit below roughly 25–45 reactivations—our two positive cells sit at 27 and 20, which is why their seed intervals are read as implementation precision, not event-sampling error. A book with a handful of reactivations gets a sign and a magnitude to price with, not significance.
(5) **Price it.** Apply the boxed rule of Section 2.4 with your sleeve size, measured dormancy scale, and honest carrying cost.
(6) **Decide, and re-estimate on a trigger.** $\hat{\Gamma}_{\text{net}}$ significantly positive and the rule satisfied: retention pays, and any isolation-preserving implementation collects it. Otherwise: do not buy retention machinery this era. Re-run on a concrete trigger, not a judgment call: whenever a new qualifying reactivation enters the inventory, and at least annually if the labeler's crisis occupancy over a rolling five years drifts from the calibration era—the paying window's location moved across eras in our record, and crisis-density drift is the covariate it moved with.

6.2 Limitations, register, and what we do not claim

Single histories. Every real-data cell is one realized market history; seed intervals quantify implementation noise conditional on that history. The record spans 1926–2025 and scores in both directions, but it is three walk-forward windows and one country.

Aggregation. The equity substrate is industry portfolios; aggregation itself can manufacture slow dynamics [19], crisis-conditional behavior is a documented feature of this territory [4, 5, 15, 32], and industry-level effects need not survive disaggregation [10]. The lodged CRSP constituent test is the decider, and until it resolves, the deficit is a fact about the aggregate cross-section.

One labeler family, one era-dependent window. The positive deficits live under drawdown- and recession-type labelers with slow crisis dynamics; the paying threshold relocated across eras. The protocol ships with the instruction to estimate the window locally, not with a universal constant.

Mechanism. On real data the measured deficit is a persistent allocation-level service gap, not demonstrated forgetting; the controlled dissociation of mechanisms lives in the companion [2], and buying our diagnostic does not require buying any mechanism story.

No alpha. Every number here is decision regret at portfolio level, gross except where a modeled crisis-window cost schedule is stated; nothing is a claim of live trading profitability [23]. The protocol's product is a decision—pay to remember, or don't—and its warranty is the record of Table 2: three flats, two inversions, two orderings, all published under the same rule.

References

[1] Andrew Ang and Allan Timmermann. 2012. Regime changes and financial markets. *Annual Review of Financial Economics* 4, 1 (2012), 313–337.
[2] Anonymous. 2026. When the Regime Returns: Retention and Invariance in Decision-Focused Strategy Pools for Non-Stationary Markets. Companion manuscript, in preparation.
[3] David H Bailey and Marcos López de Prado. 2014. The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management* 40, 5 (2014), 94–107.
[4] Michael J. Cooper, Roberto C. Gutierrez, and Allaudeen Hameed. 2004. Market States and Momentum. *Journal of Finance* 59, 3 (2004), 1345–1365.
[5] Kent Daniel and Tobias J. Moskowitz. 2016. Momentum Crashes. *Journal of Financial Economics* 122, 2 (2016), 221–247.
[6] Darrell Duffie. 2010. Presidential Address: Asset Price Dynamics with Slow-Moving Capital. *Journal of Finance* 65, 4 (2010), 1237–1267.
[7] Graham Elliott and Allan Timmermann. 2005. Optimal Forecast Combination Under Regime Switching. *International Economic Review* 46, 4 (2005), 1081–1102.
[8] Adam N Elmachtoub and Paul Grigas. 2022. Smart "predict, then optimize". *Management Science* 68, 1 (2022), 9–26.
[9] Amit Goyal and Ivo Welch. 2008. A Comprehensive Look at the Empirical Performance of Equity Premium Prediction. *Review of Financial Studies* 21, 4 (2008), 1455–1508.
[10] Bruce D. Grundy and J. Spencer Martin. 2001. Understanding the Nature of the Risks and the Source of the Rewards to Momentum Investing. *Review of Financial Studies* 14, 1 (2001), 29–78.
[11] Massimo Guidolin and Allan Timmermann. 2007. Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control* 31, 11 (2007), 3503–3544.
[12] James D Hamilton. 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57, 2 (1989), 357–384.
[13] Peter R Hansen. 2005. A test for superior predictive ability. *Journal of Business & Economic Statistics* 23, 4 (2005), 365–380.
[14] Campbell R Harvey and Yan Liu. 2016. Backtesting. *The Journal of Portfolio Management* 42, 1 (2016), 13–28.
[15] Sam James Henkel, J. Spencer Martin, and Federico Nardari. 2011. Time-Varying Short-Horizon Predictability. *Journal of Financial Economics* 99, 3 (2011), 560–580.
[16] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. 2017. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences* 114, 13 (2017), 3521–3526.
[17] Saurabh Kumar, Henrik Marklund, Ashish Rao, Yifan Zhu, Hong Jun Jeon, Yueyang Liu, and Benjamin Van Roy. 2025. Continual Learning as Computationally Constrained Reinforcement Learning. *Foundations and Trends in Machine Learning* 18, 5 (2025), 913–1053. arXiv:2307.04345.
[18] Juhani T. Linnainmaa and Michael R. Roberts. 2018. The History of the Cross-Section of Stock Returns. *Review of Financial Studies* 31, 7 (2018), 2606–2649.
[19] Andrew W. Lo and A. Craig MacKinlay. 1988. Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test. *Review of Financial Studies* 1, 1 (1988), 41–66.
[20] Marcos López de Prado. 2018. *Advances in Financial Machine Learning*. John Wiley & Sons.
[21] David Lopez-Paz and Marc'Aurelio Ranzato. 2017. Gradient Episodic Memory for Continual Learning. In *Advances in Neural Information Processing Systems 30.*

[22] Jayanta Mandi, James Kotary, Senne Berden, Maxime Mulamba, Victor Bucarey, Tias Guns, and Ferdinando Fioretto. 2024. Decision-focused learning: Foundations, state of the art, benchmark and future opportunities. *Journal of Artificial Intelligence Research* 80 (2024), 1623–1701.

[23] R David McLean and Jeffrey Pontiff. 2016. Does academic research destroy stock return predictability? *The Journal of Finance* 71, 1 (2016), 5–32.

[24] Mark Mitchell, Lasse Heje Pedersen, and Todd Pulvino. 2007. Slow Moving Capital. *American Economic Review* 97, 2 (2007), 215–220.

[25] Tobias J. Moskowitz and Mark Grinblatt. 1999. Do Industries Explain Momentum? *Journal of Finance* 54, 4 (1999), 1249–1290.

[26] Peter Nystrup, Petter N Kolm, and Erik Lindström. 2020. Greedy online classification of persistent market states using realized intraday volatility features. *The Journal of Financial Data Science* 2, 3 (2020), 25–39.

[27] Peter Nystrup, Henrik Madsen, and Erik Lindström. 2018. Dynamic portfolio optimization across hidden market regimes. *Quantitative Finance* 18, 1 (2018), 83–95.

[28] Chaofan Pan, Lingfei Ren, Linbo Xiong, Yonghao Li, Wei Wei, and Xin Yang. 2026. ReCAP: Regime-Adaptive Continual Learning for Portfolio Management. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. arXiv:2606.00143.

[29] German I Parisi, Ronald Kemker, Jose L Part, Christopher Kanan, and Stefan Wermter. 2019. Continual lifelong learning with neural networks: A review. *Neural Networks* 113 (2019), 54–71.

[30] M. Hashem Pesaran and Allan Timmermann. 2007. Selection of Estimation Window in the Presence of Breaks. *Journal of Econometrics* 137, 1 (2007), 134–161.

[31] Daniel Philps, Tillman Weyde, Artur d'Avila Garcez, and Roy Batchelor. 2018. Continual learning augmented investment decisions. *arXiv preprint arXiv:1812.02340* (2018). NeurIPS 2018 Workshop on Challenges and Opportunities for AI in Financial Services.

[32] David E. Rapach, Jack K. Strauss, and Guofu Zhou. 2010. Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real Economy. *Review of Financial Studies* 23, 2 (2010), 821–862.

[33] David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy P. Lillicrap, and Greg Wayne. 2019. Experience Replay for Continual Learning. In *Advances in Neural Information Processing Systems 32*.

[34] Halbert White. 2000. A reality check for data snooping. *Econometrica* 68, 5 (2000), 1097–1126.